

🏆 HACK THE LAW 2026 · LEGORA CHALLENGE

The Sparring Room

A voice-first legal-negotiation trainer.

Not an assistant. A sparring partner.

TEAM SILK

VOICE · CLAUDE · NEMOTRON

NEXT.JS 16

The apprenticeship is thinning.

Juniors used to learn by doing the grunt work — drafting, marking up, arguing small points. As AI absorbs that work, the **reps that built judgement disappear.**

THE GAP

You can't learn to negotiate by reading.

- No safe place to argue out loud and lose
- No opponent that **holds its ground** instead of agreeing
- No immediate, grounded feedback on what you missed

Deliberate practice against genuine resistance.

01

Negotiate out loud

Real-time spoken argument against an AI opposing counsel that doesn't fold.

02

Stay coherent

An explicit game-state layer tracks positions & concessions — it never forgets what was agreed three turns ago.

03

Get coached

At round end the AI flips to coach — scores you against a model playbook, names the turning point, offers a rematch.

A well-defined task · immediate feedback · unlimited reps.

Marlin Health Trust v Nemo

A **Data Processing Agreement** for running children's health records through an AI diagnostics company. You act for Marlin — the data controller — and defend the GDPR clauses.

GDPR ART 28(3)

ART 9(2)

BREACH NOTIFICATION

THE TRAP

A baited "Warranties" clause

One clause is deliberately weak. The adversary asks "*why did you pick this?*" — rewarding juniors who spot it, exposing those who don't. Gold-standard answers are graded behind the scenes and **never shown** during the fight.

One mounted voice session, a seven-step wizard.



The whole UI is one client component (`app/page.tsx`) wrapped in a single ElevenLabs `<ConversationProvider>` so the live voice session stays mounted across every step.

GO AGAIN ↺ ONE-CLICK REMATCH

JUNIOR · PLAYS

The trainee

- Picks case, opponent and clauses
- Argues a live round — voice or typed
- Reads the scored debrief, hits

Go Again

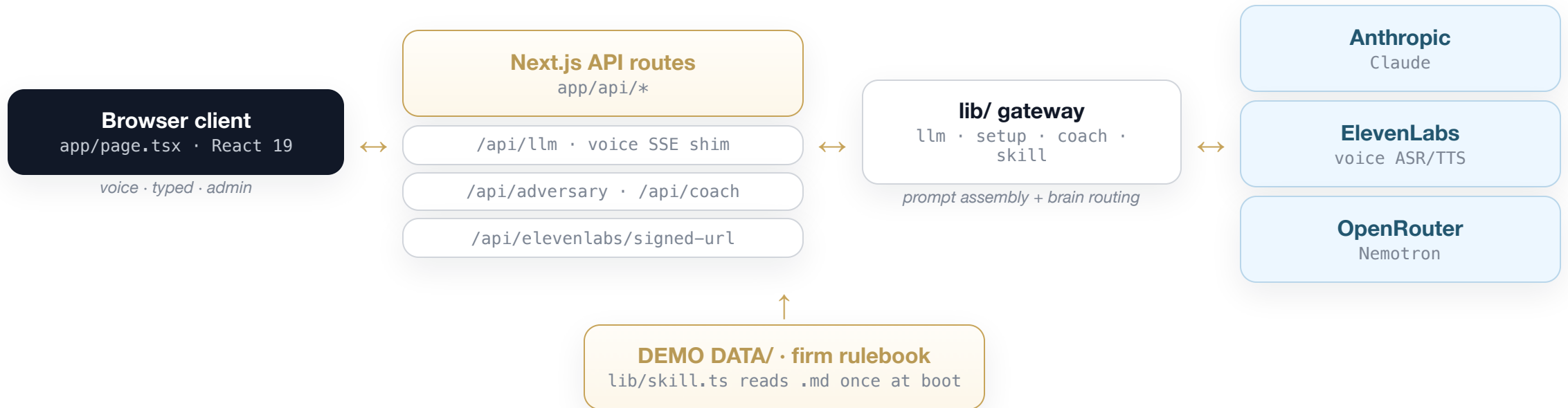
ADMIN · AUTHORS

The senior lawyer

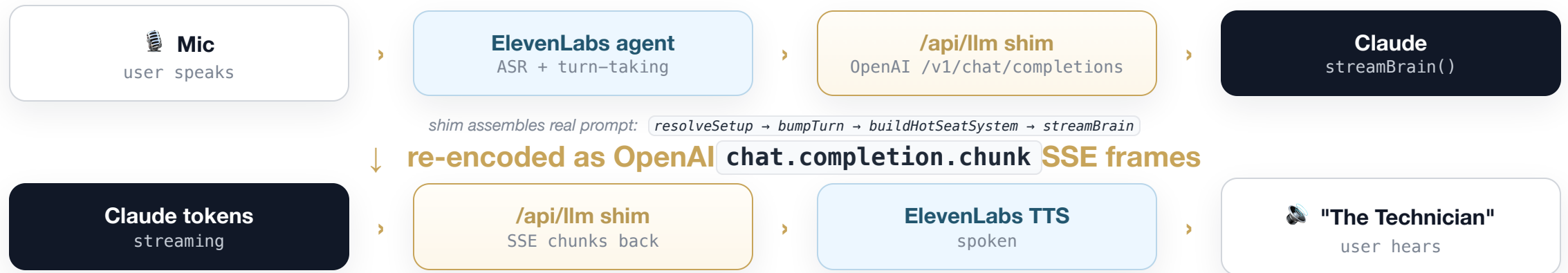
- Uploads a firm playbook → simulated build
- Tests the generated game, or re-uploads
- Amend chat: *"make the opponent more aggressive"*

① Admin upload & amend are **Wizard-of-Oz** today — framework & UX are real; document-driven generation is the next seam.

One Next.js app · three external brains · a read-only rulebook.



ElevenLabs thinks it's calling OpenAI. It's actually calling Claude.



An OpenAI-compatible translation layer.

The shim speaks OpenAI on the wire and Anthropic underneath — so any Custom-LLM voice platform plugs straight in.

- **Format translation**

`lib/openaiCompat.ts` maps OpenAI chat $\rightleftharpoons$ Claude messages, enforces Anthropic's "must start with a user turn" rule.

- **Barge-in & disconnect safe**

handles the client cutting in or dropping mid-stream without orphaning the Claude call.

- **Effectively stateless**

ElevenLabs resends full history each turn, so `lib/gameState.ts` only counts turns — a documented swap point for a Neo4j argument graph.

- **Key stays server-side**

`/api/elevenlabs/signed-url` mints the WebSocket URL so the ElevenLabs key never reaches the browser.

Cloud or Sovereign — same prompt, swappable engine.

lib/llm.ts · brain gateway
streamBrain() / generateBrain()



CLOUD · DEFAULT

Claude

@anthropic-ai/sdk · claude-sonnet-4-6

Prompt-cached rulebook (cache_control: ephemeral), thinking: disabled , low effort.

LOCAL · SOVEREIGN

NVIDIA Nemotron

openai SDK → OpenRouter · nemotron-3-super-120b · strips <think> reasoning.

Graceful fallback to Claude (fallback: true) if Local is down · UI EngineToggle.tsx flips engine and a dark green-neon theme.

The AI argues from the firm's own playbook.

`lib/skill.ts` loads the firm's markdown — method, DP checklist, gold-standard playbook — **once at boot**. It's the source of truth for how the AI questions and grades.

Drop any `.md` into `DEMO DATA/` and it's auto-appended to the rulebook.

buildHotSeatSystem()

`lib/setup.ts`

splits into



Rulebook block

`cached` · `cache_control`



Instructions

`volatile` · `per-turn setup`

Big stable prefix is cached; only the small volatile tail changes per turn → fast, cheap streaming.



Gold standards are NEVER shown to the user during the fight — only the coach reveals them, after.

It only asks questions.

A live Socratic interrogator: short, sharp turns; never reveals the gold standard; always pushes a legal point through to its **commercial consequence**. Holds its pushback defences — it does not fold.

[prompts/hotseat.ts](#)

Persona × Difficulty

OPPOSING COUNSEL

STONEWALLER

CHARMER

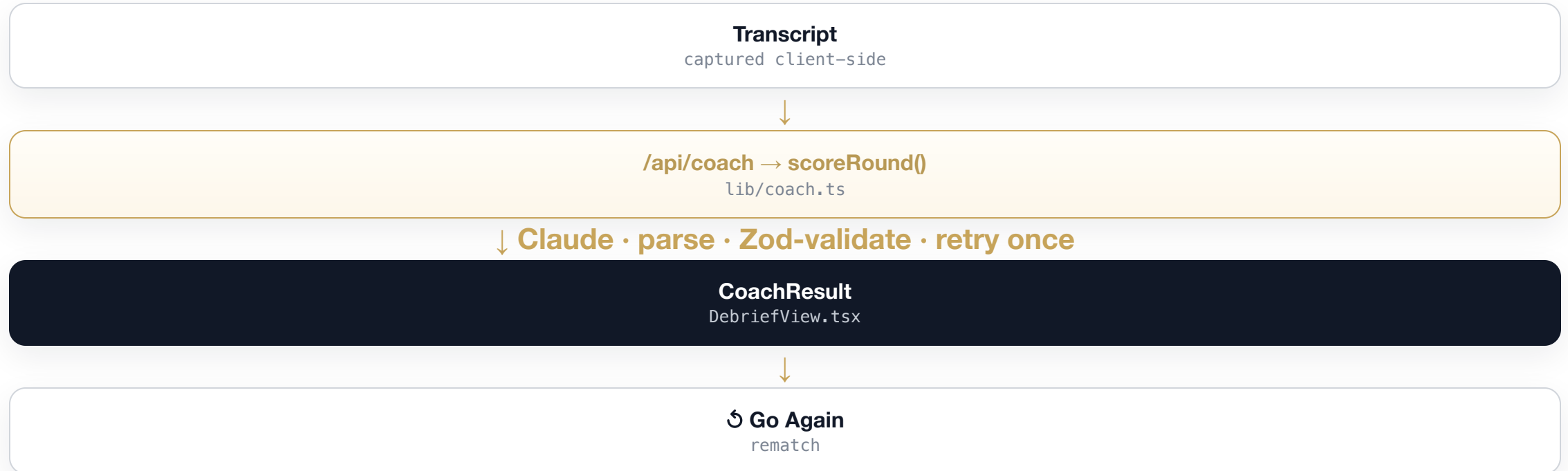
ASSOCIATE

PARTNER

SHARK

Persona & difficulty modifiers are appended to the adversary prompt — same brain, different temperament and ferocity. Voice runs through `/api/llm`; the typed fallback uses `/api/adversary` with no tunnel needed.

A scored, Socratic debrief.



What the card shows

- **0–100 health score**
graded vs the rulebook's gold-standard answers
- **Four-part consequences**
civil · GDPR · financial · reputational
- **Gaps & faulty assumptions**
issue → legal → commercial, traps picked

Lean, modern, voice-native.

16.2.9

Next.js

App Router · the "non-standard" v16 `AGENTS.md` warns about

19.2.4

React

Single stateful client component, hooks only

5.x

TypeScript

Strict · `@/*` path alias

^0.106

Anthropic SDK

Claude — adversary & coach · prompt caching

^1.9

@elevenlabs/react

Conversational AI — ASR · TTS · turn-taking

^6.45

openai SDK

Pointed at OpenRouter → NVIDIA Nemotron

v4

Tailwind

CSS-first · tokens in `globals.css`

^4.4

Zod

Validates the structured coach output

An institutional Greek theme — that goes dark in Sovereign mode.

CLOUD · DEFAULT

Marble & gold

 Greek gold `#c7a45a`

 Ink `#111827` · serif headings (Playfair)

Greek-hall hero · mic orb · voice visualizer

LOCAL · SOVEREIGN

Black & green-neon

 Neon green `#4ade80`

 Near-black `#070d0a`

Same layout — `.engine-dark` overrides only the tokens

The visual identity is the engine story: Cloud feels institutional; the Sovereign/Local brain literally darkens the room.

Every placeholder is a labelled seam.

STATE

Neo4j argument graph

Replaces the turn-counter in `gameState.ts` for true memory.

GROUNDING

EU Cellar API

Live GDPR text via `fetchGrounding()` in `data/grounding.ts`.

RESEARCH

Perplexity Sonar

Unused `research` role already wired in the gateway.

SCORING

Independent judge

Swap the coach for a separate Nemotron grader.

Next big step: document-driven admin generation — turn an uploaded firm playbook into a real, custom game instead of the Wizard-of-Oz placeholder.

— THE SPARRING ROOM

Voice in. Resistance back. Judgement built.

A grounded AI opponent that holds its ground, a coach that scores you against the firm's own playbook, and a single tap to *go again*.

ELEVENLABS VOICE

CLAUDE · NEMOTRON

NEXT.JS 16 · REACT 19

ZOD-GRADED DEBRIEF